

1. Objectives:

1. To understand the concept of Class and Object in OOP.
2. To understand how Constructor works in OOP.
3. To learn how Static keyword works in OOP.

2. Introduction:

OOP: A **Class** is a blueprint or template that defines the properties (variables) and behaviors (methods) of an entity. An **object** is an instance of a class, which represents a real-world entity and contains actual values of those properties. Objects are created using constructors and are used to access the data and methods defined in a class.

Constructor: It is a special method in a class that is automatically called when an object is created. Its main purpose is to initialize the values of object variables. A **method** is a function defined inside a class that performs a specific task, such as displaying or processing data. Methods allow code reuse and help organize the program logically.

Static: It is a class-level variable that is shared among all objects of a class. Unlike instance variables, static variables do not create separate copies for each object. They are commonly used to store common information or to keep track of the number of objects created.

3. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** Antigravity

4. Experiment Implementation:

Experiment :

Main Class :

```
package FifthLab;

public class MainClass {

    public static void main(String[] args) {

        CSE c1 = new CSE(1024, "Rezwan", 22, "Male");
        c1.info();
        System.out.println();

        CSE c2 = new CSE(1015, "Avoy", 21, "Male");
        c2.info();
        System.out.println();

        CSE c3 = new CSE(1004, "Soccho", 20, "Male");
        c3.info();
        System.out.println();

        System.out.println("Students of CSE Called: " + CSE.count);

    }

}
```

CSE Class :

```
package FifthLab;

public class CSE {

    static String uni = "BAUST";
    static String dep = "CSE";
```

```

static int count = 0;

int id;
String name;
int age;
String gender;

CSE(int i, String n, int a, String g) {
    id = i;
    name = n;
    age = a;
    gender = g;
    count++;
}

void info() {
    System.out.println("Department: " + dep);
    System.out.println("ID: " + id);
    System.out.println("Name: " + name);
    System.out.println("Age: " + age);
    System.out.println("Gender: " + gender);
    System.out.println("University: " + uni);
}
}

```

3.1: Experiment 1 output

```

PS D:\BAUST CSE\Java Programming\5th_Lab\5thLab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-XX:+ShowCodeDetailsIn
BAUST CSE\Java Programming\5th_Lab\5thLab\bin' 'FifthLab.MainClass'
Department: CSE
ID: 1024
Name: Rezwan
Age: 22
Gender: Male
University: BAUST

Department: CSE
ID: 1015
Name: Avoy
Age: 21
Gender: Male
University: BAUST

Department: CSE
ID: 1004
Name: Soccho
Age: 20
Gender: Male
University: BAUST

Students of CSE Called: 3
PS D:\BAUST CSE\Java Programming\5th_Lab\5thLab>

```

5. Discussion & Conclusion:

In this lab, the concept of OOP was implemented using classes and objects in Java. Two different classes were created to represent different departments, and objects were instantiated using constructors. The info() method was used to display the details of each object. Additionally, a static variable was used to count the number of objects created, which helped in understanding shared data among objects. Through this lab, the use of constructors, methods, and static variables became clear, and it improved the understanding of basic object-oriented programming concepts in Java.

7. Reference:

1. <https://www.geeksforgeeks.org/java/inheritance-and-constructors-in-java/> accessed at 4.20 PM on 24th April 2026.
2. <https://www.geeksforgeeks.org/java/static-keyword-java/> accessed at 6.25 PM on 24th April 2026.